BRAC UNIVERSITY

Inspiring Excellence

CSE370 : Database Systems
Project Report
Project Title : Criminal Analysis and Investigation System

Group No : 09, CSE370 Lab Section : 02, Spring 2026		
ID	**Name**	**Contribution**
23301563	Kumar Shagata Biswas	EER Diagram, Frontend, Backend, Data Base design, 2 features,Lab report
23201118	Abir Enam	Scheema , Lab Report, Frontend,Backend,2 features,
23201047	Daud Ibrahim Hassan	Normalization, Database design, Lab report, Frontend,2 features, Backend

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Introduction

A robust, law-enforcement web application built to track criminal profiles, manage investigations, log evidence, and identify crime patterns using advanced data relationships.

The main system is built around a database that reflects how investigations actually happen in the real world. It helps the police department to piece together every part of information such as criminal records, officer details, victim data, crime reports and evidence. It makes the department easy to connect structure, making it easier to store, access and analyze relationships in one place.

Each part of the date is thoughtfully linked and it allows the officers to see the full picture of the case rather than piercing them together.Whether it's tracking which officers worked on a crime, identifying repeat offenders, or connecting victims to incidents, the relationships between the data help uncover patterns that might otherwise go unnoticed.

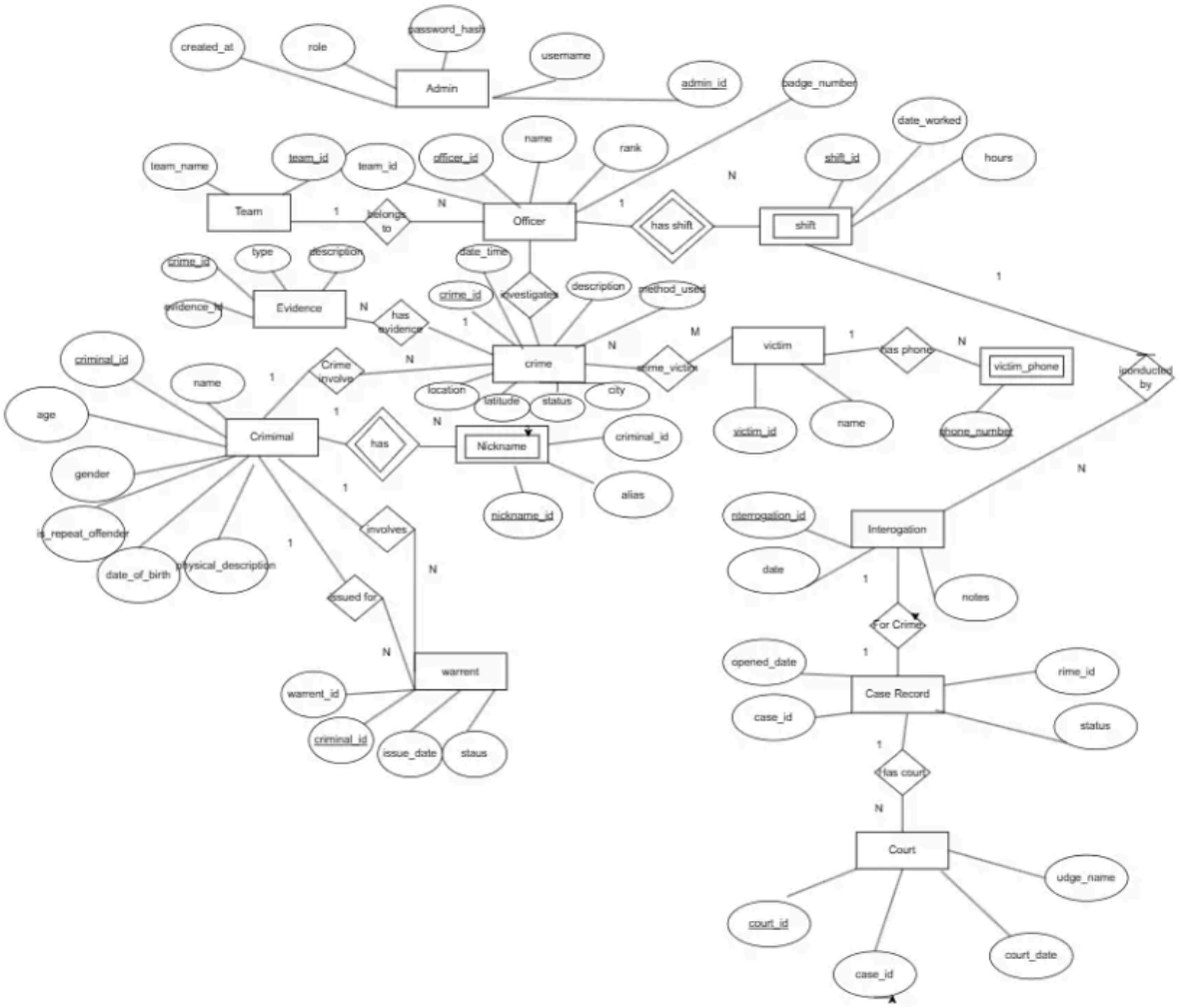
The system makes practical, day to day operations like managing teams, recording stuff, keeping the case records up to date and logging evidence. It was built to optimize search for certain items and special views that highlight important insights—like crime trends or wanted individuals; it helps turn large amounts of data into meaningful information.

In conclusion we can say that this database is the core part for officers and crime investigators to help and enforce law, maintain cases and information, and keep records of the criminals. Keeping the record of the victims and the time and pattern of the crime. Time table of the crimes and linking everything together so that any information can be extracted at any certain time.

Project Features

ID, Name	Features [3 per member]	
23301563, Kumar Shagata Biswas	Ft 1	Officer Directory
	Ft 2	Team Management
	Ft 3	Warrant Tracking
23201118, **Abir Enan**	Ft 1	Crime Records
	Ft 2	Nickname Tracking
	Ft 3	Victim Rooster
23201047 Daud Ibrahim	Ft 1	Criminal Registry
	Ft 2	Crime Pattern Analysis
	Ft 3	Be On The Lookout

ER/EER Diagram



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Schema Diagram

https://drive.google.com/file/d/1Dm4NTtBR7OZgHF8w5OawQ1hWJY3A2Xx-/view

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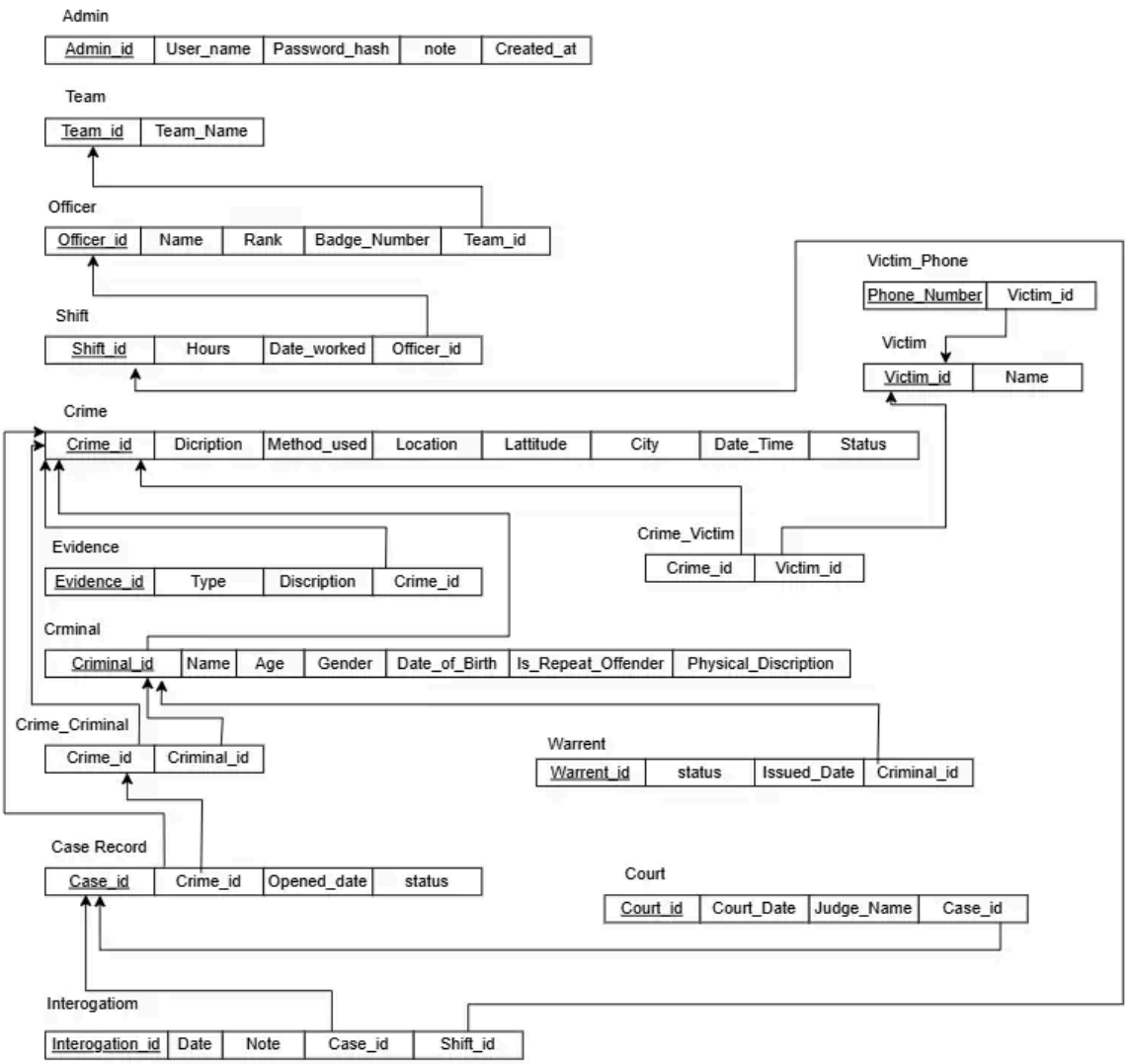


Normalization

a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.
b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?
c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?

Admin (Admin_id, User_name, Password_hash, Note, Created_at)
Team (Team_id, Team_Name)
Officer (Officer_id, Name, Rank, Badge_Number, Team_id)
Shift (Shift_id, Hours, Date_worked, Officer_id)
Crime (Crime_id, Description, Method_used, Location, Latitude, City, Date_Time, Status)
Evidence (Evidence_id, Type, Description, Crime_id)
Criminal (Criminal_id, Name, Age, Gender, Date_of_Birth, Is_Repeat_Offender, Physical_Description)
Crime_Criminal (Crime_id, Criminal_id)
Victim (Victim_id, Name)
Victim_Phone (Phone_Number, Victim_id)
Crime_Victim (Crime_id, Victim_id)
Case_Record (Case_id, Crime_id, Opened_date, Status)
Court (Court_id, Court_Date, Judge_Name, Case_id)
Warrant (Warrant_id, Status, Issued_Date, Criminal_id)
Interrogation (Interrogation_id, Date, Note, Case_id, Shift_id)

a) This is the schema table. And we can see that each table is already in 1NF form. There is no multivalued attribute, no nested relations and no composite attributes in any of the tables stated above. Multi-valued attributes like phone numbers are already separated into Victim_phone with one phone per row; this is correct 1NF handling. And many to many relationships are already solved via junction tables

b) The 2NF is already done because all the tables have a single primary key. So there is no chance that one partial key will define an independent attribute. As a result all the tables must be in 2NF way. No, there can not be any partial

functional dependencies in your relational schema. Here 2 tables Crime_Criminal and Crime_Verdict table have 2 primary keys but no attributes are partially dependent on it. So it is normally in 2NF form.

c) We can see 2 tables Crime and Court violate the 3NF. So the reasons why they are not in 3NF form is :

For Crime: Crime_id → City → City_Name
For Court: Court_id → Judge_id → Judge_Name

To transform it in 3NF:
City (City_id, City_Name)
Crime (Crime_id, Description, Method_used, Location, Latitude, City_id, Date_Time, Status)

Judge (Judge_id, Judge_Name)
Court (Court_id, Court_Date, Judge_id, Case_id)

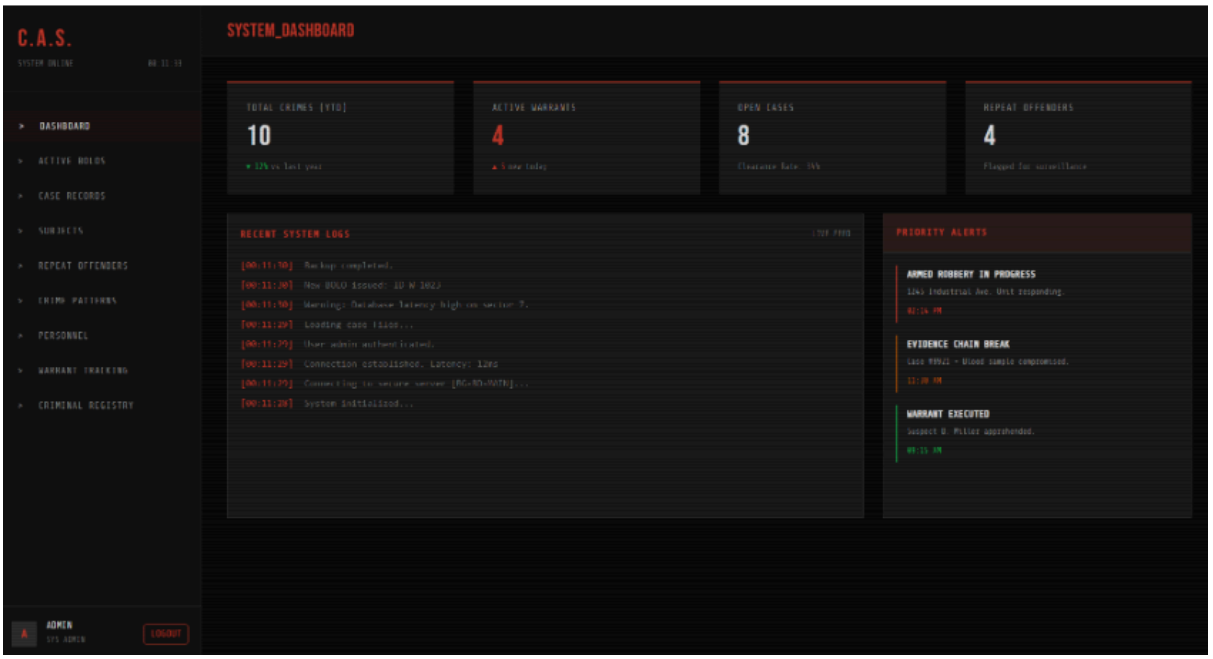
So by decomposing the table like this we will make the whole schema Normalized in 3NF form

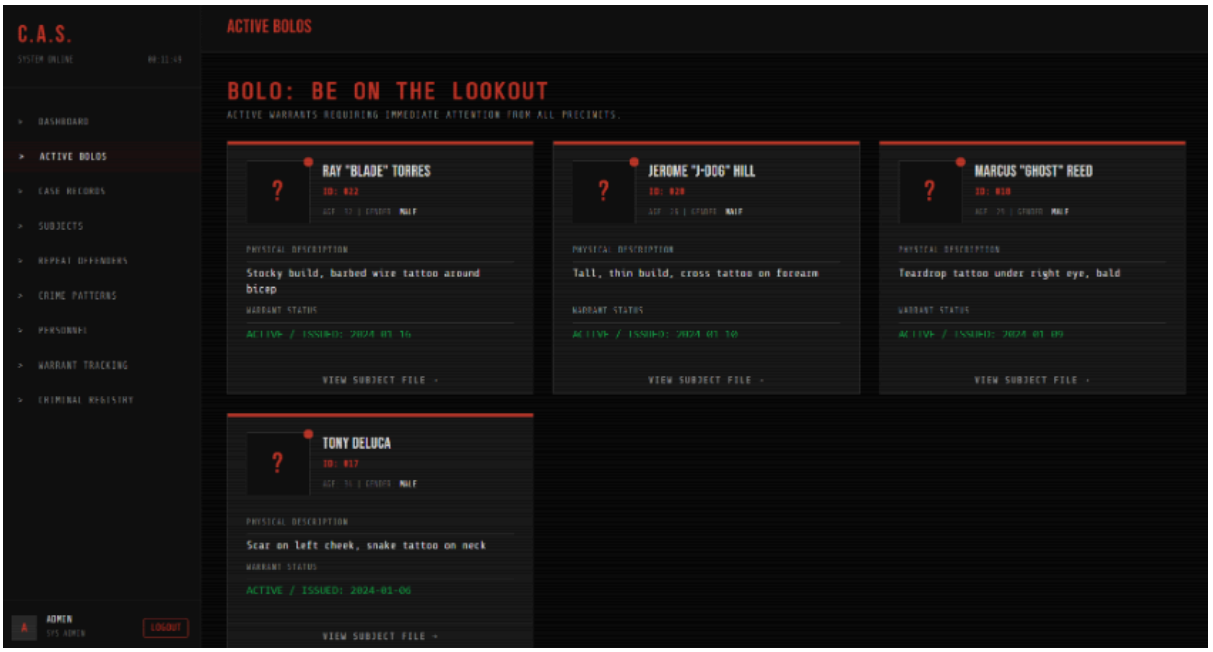
Frontend Development

Briefly discuss about Backend Development and add relevant Screenshots (if required) by mentioning Individual Contributions

Contribution of ID : 23301563, Name : Kumar Shagata Biswas

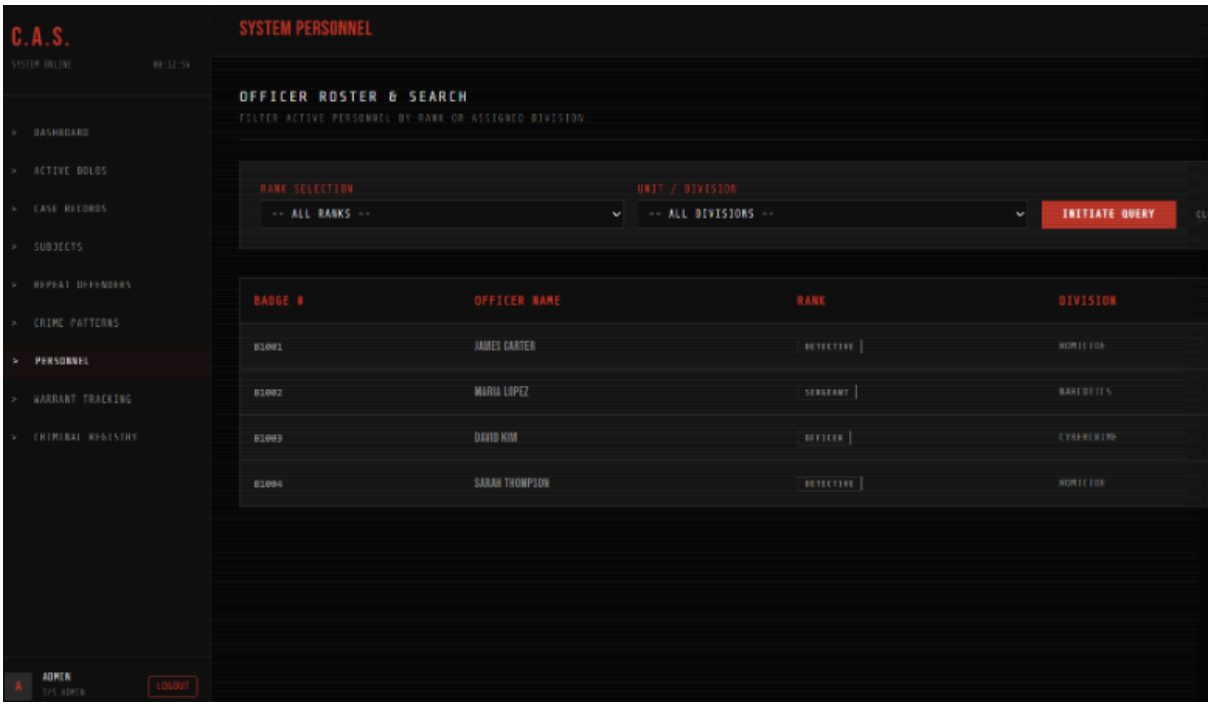
Created basic page structures using HTML
Added simple styling using Tailwind CSS
Organized content layout for different pages

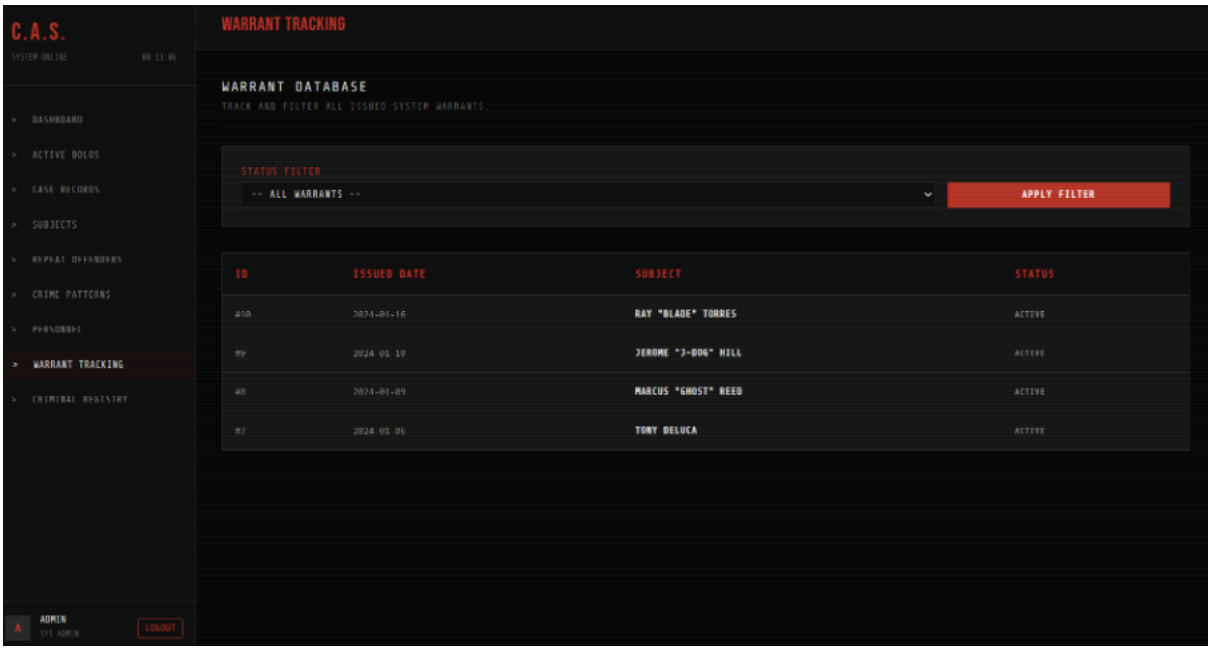




Contribution of ID : 23201118**, Name :Abir Enam**

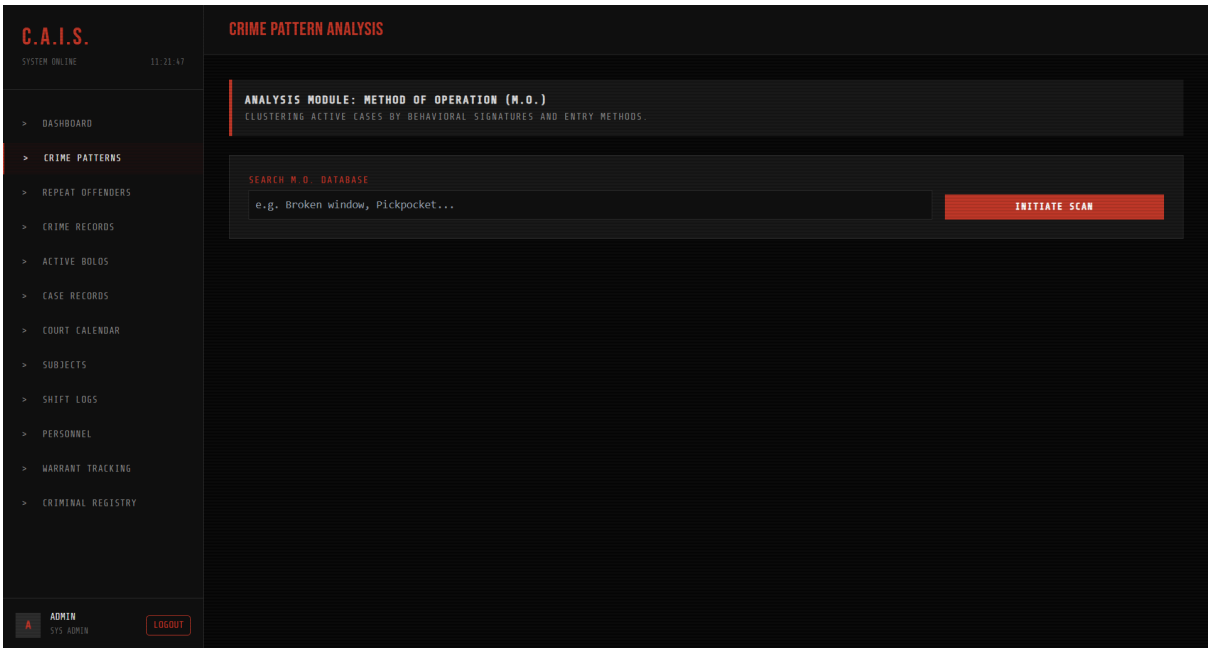
Developed interactive features using Vanilla JavaScript
Handled form validations and dynamic user interactions
Assisted in integrating frontend with backend functionality

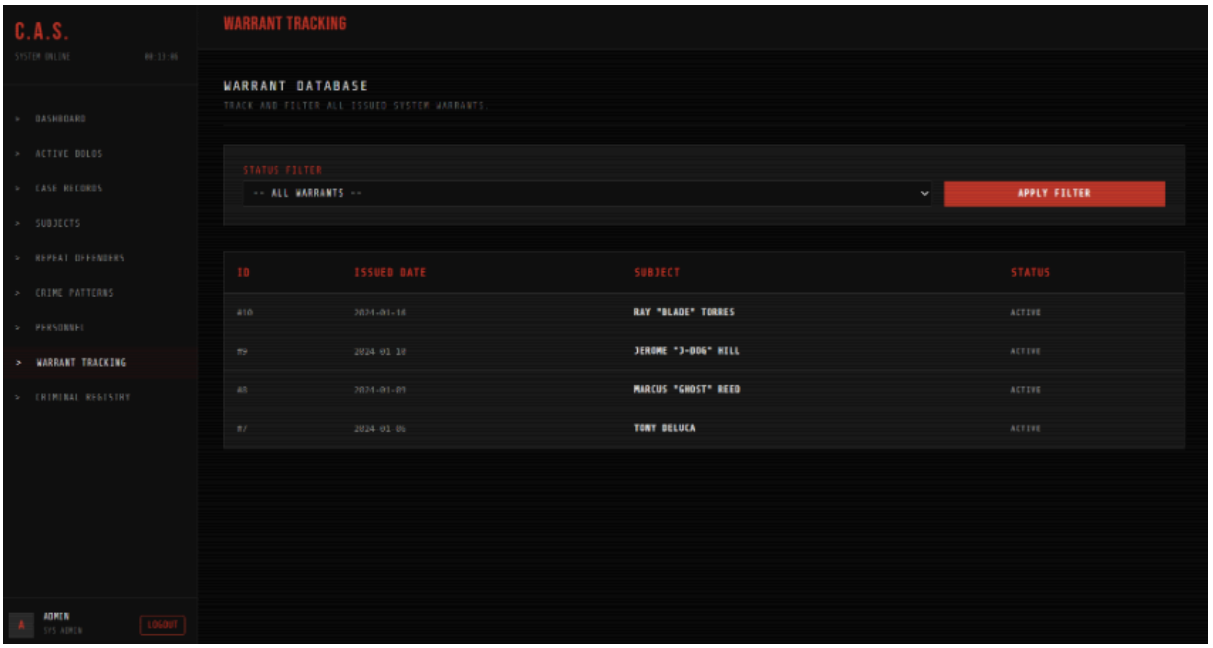




Contribution of ID : 23201047, Name : Daud Ibrahim

Worked on UI styling and responsiveness using Tailwind CSS
Improved design consistency across pages
Assisted in testing and debugging frontend components





Backend Development

Briefly discuss about Backend Development and add relevant Screenshots (if required) by mentioning Individual Contributions

Contribution of ID : 23301563, Name : Kumar Shagata Biswas

Case Lifecycle:

Managed case status such as “Open” and “Finished”
Allowed updating case progress during investigation
Helped track the current state of each case

Warrant Tracking :

Stored warrant details linked to criminals
Tracked warrant status as Active or Cancelled
Helped identify currently wanted individuals

Officer Directory:

Stored officer details like name, rank, and badge number
Allowed basic CRUD operations for officer management

Contribution of ID : 23201118, Name : Abir Enan

Team Management:

Assigned officers to different teams
Maintained relationships between officers and teams

Victim Roster :

Stored victim information and contact details
Maintained relationship between victims and crimes

Crime Records:

Logged crime details such as location, time, and method
Maintained records for investigation purposes

Contribution of ID : 23201047, Name : Daud Ibrahim

Crime Pattern Analysis:

Collected and analyzed crime data stored in the database
Identified patterns based on crime type, location, and time

Criminal Registry:

Stored criminal profiles including personal details
Allowed updating and managing criminal records

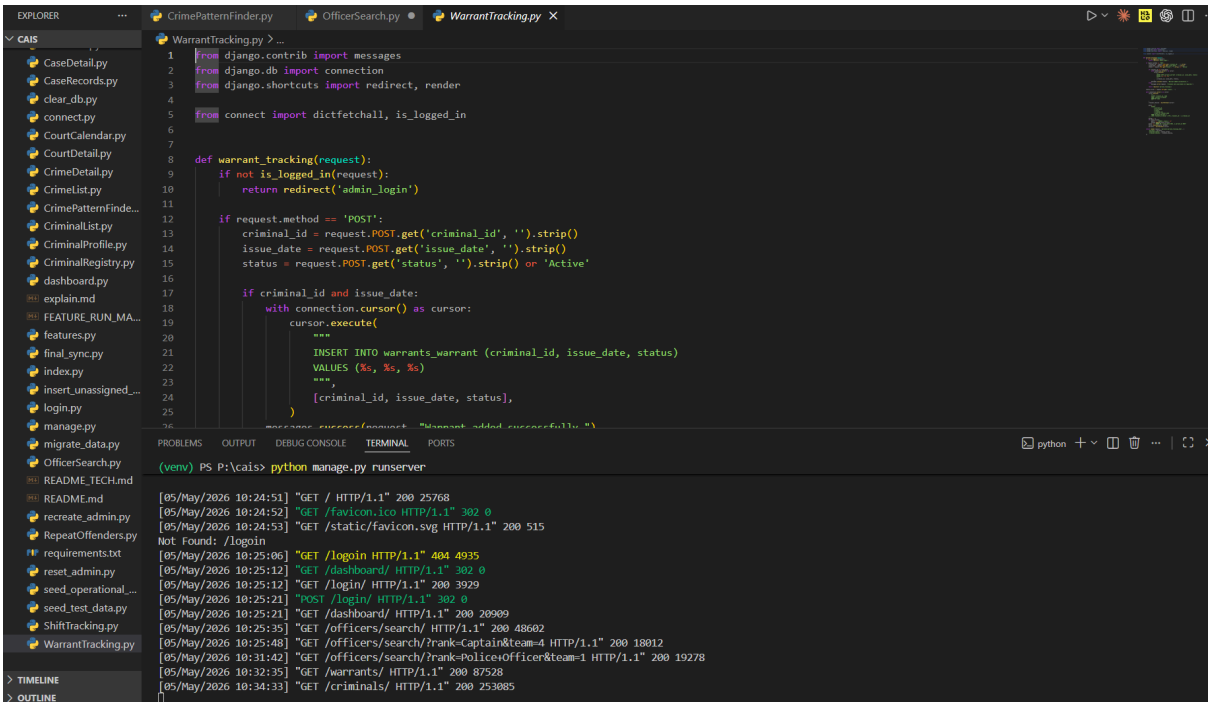
Admin Authentication :

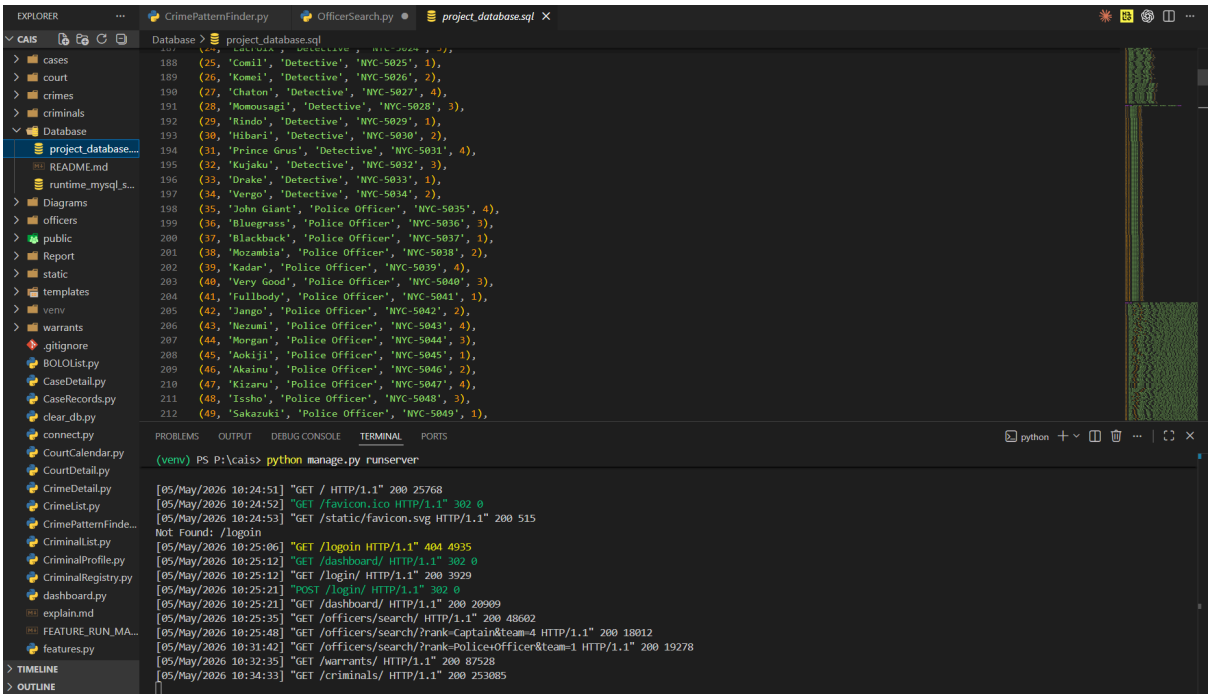
Implemented basic login system using stored admin credentials
Verified user identity before allowing system access
Ensured only authorized users can access the system

Source Code Repository

Upload source code to GitHub or Google Drive and share a publicly accessible link in this section of the report.

https://github.com/daudibrahimhasan/CSE370-Project-CAIS





Conclusion

The Criminal Analysis & Investigation System (CAIS) successfully demonstrates how a structured database and web application can support modern law enforcement operations. The system efficiently manages criminal records, officer details, crime reports, evidence, and case progress through well-defined relationships and organized data storage.

Through the implementation of both frontend and backend components, the project provides a user-friendly interface along with reliable backend functionality. Features such as warrant tracking, crime pattern identification, and repeat offender alerts highlight how data can be transformed into meaningful insights.

Overall, this project enhances our understanding of database design, relational modeling, and full-stack development. It also shows how technology can be used to improve efficiency, accuracy, and decision-making in real-world investigative scenarios.

References

MySQL link : https://dev.mysql.com/doc/
W3Schools SQL Tutorial: https://www.w3schools.com/sql/
Tailwind CSS Documentation : https://tailwindcss.com/docs
MDN Web Docs (HTML, CSS, JavaScript): https://developer.mozilla.org/
Drrible: https://dribbble.com/

Course Materials (CSE370 Database Systems)
https://www.google.com/search?q=bux&oq=bux&gs_lcrp=EgZjaHJvbWUyDwgAEEUYORiDARixAxiABDIHCAEQABiABDIHCAIQABiABDIHCAMQABiABDIHCAQQABiABDIHCAUQABiABDIHCAYQABiABDIHCAcQABiABDIHCAgQLhiABDIHCAkQABiABNIBCTIxOTJqMGoxNagCCLACAfEFsRmzJF17T9vxBbEZsyRde0_b&sourceid=chrome&ie=UTF-8